Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-II-Q03] | Part A (Q1-Q5)

Answer: 021

1. The pyramid height is $h = 3 \cdot 54/36 = 9/2$. Its sphere center lies on the symmetry axis, at signed height z above the base.

2. The squared distance to a base vertex is $z^2 + 18$; to the apex it is $(9/2 - z)^2$. Equating yields $z = 1/4$.

3. The radius is $9/2 - 1/4 = 17/4$, hence $m + n = 21$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q03] | Part A (Q1-Q5)

Answer: 079

1. If P is distance d from the disk center, the small-sphere center is height 42, at distance $\sqrt{d^2+42^2}$ from the hemisphere center. Containment requires this distance plus 42 at most 200.

2. Thus $d^2 \le 158^2 - 42^2 = 23200$. Allowed points form a disk, so the area ratio is $23200/40000 = 29/50$, giving 79.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-I-Q06] | Part B (Q6-Q10)

Answer: 173

1. The centers have heights $\sqrt{r^2-1}$ and $\sqrt{r^2-4}$. Hence $49+(\sqrt{r^2-1}-\sqrt{r^2-4})^2=4r^2$.

2. Put $u=r^2$. Rearrangement gives $\sqrt{(u-1)(u-4)}=22-u$. Squaring cancels u^2 and yields $39u=480$.

3. Thus $u=160/13>4$ and $22-u>0$, so the unsquared equation is satisfied. The requested sum is 173.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2020-II-Q07] | Part B (Q6-Q10)

Answer: 298

1. Orient each axis from its base toward its apex, with origin O. In an axial half-plane, the side is $3u + 8\rho = 15$, since the apex is at $u = 5$ and the base edge at $(u, \rho) = (-3, 3)$.

2. For sphere center axial coordinates u, v, its distances to the axes are at least $|v|, |u|$. Thus $\sqrt{73}r \le 15 - 3u - 8|v|$ and $\sqrt{73}r \le 15 - 3v - 8|u|$. Adding yields $2\sqrt{73}r \le 30$.

3. At O, the sphere with $r = 15/\sqrt{73}$ touches each lateral surface, and $r < 3$ keeps it above both base planes. Hence the maximum square is $225/73$, giving 298.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2021-II-Q10] | Part B (Q6-Q10)

Answer: 335

1. All three spheres lie on the same side of each tangent plane. If two were on opposite sides, external tangency would force the same contact point; tangency of the third to the opposite-side sphere would force that contact point again, making its center lie on the same normal line as the same-side sphere and preventing their external tangency. Let the sector angle be 2θ. A radius-r center has transverse distance $r/\sin\theta$ from ℓ.

2. The two equal centers differ by 72 along ℓ. The small center is midway along that direction, so tangency gives $(23/\sin\theta)^2 + 36^2 = 49^2$. Thus $\sin\theta = 23/\sqrt{1105}$ and $\cos\theta = 24/\sqrt{1105}$.

3. The contact-point distance is $13\cot\theta = 312/23$, giving 335.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2022-I-Q10] | Part B (Q6-Q10)

Answer: 756

1. Let the positive sphere-center heights be u, v, w and common section radius be r. Then $u^2 = 121 - r^2, v^2 = 169 - r^2, w^2 = 361 - r^2$.

2. The first center distance is 24, so $(v-u)^2 = 24^2 - 560 = 16$. Since $v > u$, $v - u = 4$; also $(v-u)(v+u) = 48$, hence $u = 4, v = 8$. Thus $r^2 = 105$ and $w = 16$.

3. Project the 30-unit first-to-third center segment onto the plane: $AC^2 = 30^2 - (16-4)^2 = 756$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2024-II-Q08] | Part B (Q6-Q10)

Answer: 127

1. Take an axial cross-section. The generating-circle center lies 6 units from the axis, and its distance from the sphere center is $11-3=8$ for internal tangency or $11+3=14$ for external tangency.

2. The contact point lies on the line joining these centers, at distance 11 from the sphere center. Similar triangles therefore give $r_i=6\cdot 11/8=33/4$ and $r_o=6\cdot 11/14=33/7$.

3. The difference is $99/28$, so $m+n=127$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2000-II-Q12] | Part C (Q11-Q15)

Answer: 118

1. The perpendicular projection of O to the triangle plane is its circumcenter. The triangle is acute since $15^2 < 13^2 + 14^2$, so this foot is inside and gives the distance to the triangle itself.

2. Heron gives area 84, hence circumradius $R = 13 \cdot 14 \cdot 15/(4 \cdot 84) = 65/8$.

3. The plane distance is $\sqrt{20^2 - R^2} = 15\sqrt{95}/8$. Thus $m + n + k = 15 + 95 + 8 = 118$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2001-I-Q12] | Part C (Q11-Q15)

Answer: 005

1. Tangency to the three coordinate planes puts the sphere center at (r, r, r). The fourth face lies on $x/6 + y/4 + z/2 = 1$, or $2x + 3y + 6z = 12$.

2. The center-to-face distance is $(12 - 11r)/\sqrt{4 + 9 + 36} = (12 - 11r)/7$. Equating it to r gives $18r = 12$, so $r = 2/3$ and the answer is 5.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2011-I-Q13] | Part C (Q11-Q15)

Answer: 330

1. Let the nearest height be h. Projections of the three orthogonal ten-unit edges onto the vertical direction are $10-h, 11-h, 12-h$. Their squared sum is 10^2, because the squared components of a unit vertical vector sum to one.

2. Thus $(10-h)^2+(11-h)^2+(12-h)^2=100$, or $3h^2-66h+265=0$.

3. The solutions are $(33\pm\sqrt{294})/3$. Since the nearest vertex has $h<10$, choose the minus sign. The sum is $33+294+3=330$.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2016-II-Q14] | Part C (Q11-Q15)

Answer: 450

1. Let H be the triangle center and M the midpoint of AB. Both P, Q and sphere center O lie on the perpendicular axis through H. Since $OP = OQ = d$, O is the midpoint of PQ and $PQ = 2d$. Also $MH = 100\sqrt{3}$, $CH = 200\sqrt{3}$.

2. The perpendicular section through M, H, P, Q gives angles $u = PMH, v = QMH$ with $u + v = 120^\circ$. Because PQ is a sphere diameter and C lies on the sphere, PCQ is right; its altitude gives $PH \cdot QH = CH^2$. Thus $\tan u \tan v = 4$.

3. Tangent addition gives $-\sqrt{3} = (\tan u + \tan v)/(1 - 4)$, hence the tangent sum is $3\sqrt{3}$. Therefore $2d = PH + QH = 100\sqrt{3} \cdot 3\sqrt{3} = 900$, so $d = 450$. The two positive tangents exist since their sum squared exceeds four times their product.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-I-Q12] | Part C (Q11-Q15)

Answer: 161

1. Let a common tangent plane have unit normal (u, v, h) and equation $ux + vy + hz = c$. A radius-ρ sphere above $z = 0$ has center (x, y, ρ); choose the normal so tangency reads $ux + vy - c = (1 - h)\rho$. Since the second plane is distinct from $z = 0$, $h \neq 1$, making radius an affine function of its contact point.

2. Take $A = (0, 0), B = (6, 0), C = (0, 4)$. The affine radius function is $g(x, y) = 1 + x/6 + y/2$. The centroid $(2, 4/3)$ reflects across $2x + 3y = 12$ to $D = (42/13, 124/39)$.

3. Thus $r = g(D) = 122/39$, with coprime terms, giving 161. The normal can indeed be normalized with $h = -13/23$, so the distinct tangent plane exists.

Source / 来源

Section 2.9 Spheres & spatial distances

球与空间距离

[AIME-2026-II-Q12] | Part C (Q11-Q15)

Answer: 223

1. The exceptional edges $10, 18$ must be opposite; all other edges have square 250. Realize the tetrahedron as $A = (-5, 0, 12), B = (5, 0, 12), C = (0, -9, 0), D = (0, 9, 0)$. Reflection symmetries put both centers on the z axis.

2. For $S = (0, 0, s)$, equal squared distances give $25 + (12 - s)^2 = 81 + s^2$, hence $s = 11/3$. For $R = (0, 0, t)$, face planes $4y - 3z + 36 = 0$ and $12x + 5z = 0$ give $(36 - 3t)/5 = 5t/13$, so $t = 117/16$.

3. Thus $RS = 117/16 - 11/3 = 175/48$, and $m + n = 223$.

Source / 来源